EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. If $z + z^3 = 0$ then which of the following must be true on the complex plane?

(A) $\text{Re}(z) < 0$ (B) $\text{Re}(z) = 0$ (C) $\text{Im}(z) = 0$ (D) $z^4 = 1$

MCN161

2. Number of integral values of n for which the quantity $(n + i)^4$ where $i^2 = -1$, is an integer is

(A) 1 (B) 2 (C) 3 (D) 4

MCN162

3. Let $i = \sqrt{-1}$. The product of the real part of the roots of $z^2 - z = 5 - 5i$ is

(A) – 25 (B) – 6 (C) –5 (D) 25

MCN163

4. There is only one way to choose real numbers M and N such that when the polynomial $5x^4 + 4x^3 + 3x^2 + Mx + N$ is divided by the polynomial $x^2 + 1$, the remainder is 0. If M and N assume these unique values, then $M - N$ is

(A) – 6 (B) – 2 (C) 6 (D) 2

MCN164

5. Number of complex numbers z satisfying $z^3 = \bar{z}$ is

(A) 1 (B) 2 (C) 4 (D) 5

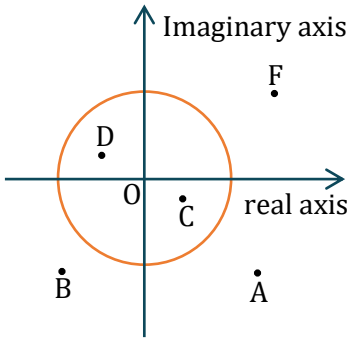
MCN165

6. Let $z = 9 + bi$ where b is non-zero real and $i^2 = -1$. If the imaginary part of z^2 and z^3 are equal, then b^2 equals

(A) 261 (B) 225 (C) 125 (D) 361

MCN166

7. The diagram shows several numbers in the complex plane. The circle is the unit circle centered at the origin. One of these numbers is the reciprocal of F, which is



(A) A (B) B (C) C (D) D

MCN167

8. If $z = x + iy$ & $\omega = \dfrac{1 - iz}{z - i}$ then $|\omega| = 1$ implies that, in the complex plane

(A) z lies on the imaginary axis (B) z lies on the real axis

(C) z lies on the unit circle (D) none

MCN168

For More Join: @IITJEE_Advanced

9. On the complex plane locus of a point z satisfying the inequality $2 \le |z-1| < 3$ denotes

(A) region between the concentric circles of radii 3 and 1 centered at (1, 0)

(B) region between the concentric circles of radii 3 and 2 centered at (1, 0) excluding the inner and outer boundaries.

(C) region between the concentric circles of radii 3 and 2 centered at (1, 0) including the inner and outer boundaries.

(D) region between the concentric circles of radii 3 and 2 centered at (1, 0) including the inner boundary and excluding the outer boundary.

MCN169

10. The complex number z satisfies $z + |z| = 2 + 8i$. The value of $|z|$ is

(A) 10 (B) 13 (C) 17 (D) 23

MCN170

11. Let $Z_1 = (8+i)\sin\theta + (7+4i)\cos\theta$ and $Z_2 = (1+8i)\sin\theta + (4+7i)\cos\theta$ are two complex numbers. If $Z_1 \cdot Z_2 = a + ib$ where $a, b \in R$ then the largest value of $(a+b)\ \forall\ \theta \in R$, is

(A) 75 (B) 100 (C) 125 (D) 130

MCN171

12. The locus of z, for $\arg z = -\frac{\pi}{3}$ is

(A) same as the locus of z for $\arg z = \frac{2\pi}{3}$

(B) same as the locus of z for $\arg z = \frac{\pi}{3}$

(C) the part of the straight line $\sqrt{3}x + y = 0$ with $(y < 0, x > 0)$

(D) the part of the straight line $\sqrt{3}x + y = 0$ with $(y > 0, x < 0)$

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MCN172

13. If z_1 & $\bar{z}_1$ represent adjacent vertices of a regular polygon of n sides with centre at the origin & if $\frac{\text{Im}\, z_1}{\text{Re}\, z_1} = \sqrt{2} - 1$ then the value of n is equal to :

(A) 8 (B) 12 (C) 16 (D) 24

MCN173

14. For $z_1 = \sqrt[6]{\frac{1-i}{1+i\sqrt{3}}}$; $z_2 = \sqrt[6]{\frac{1-i}{\sqrt{3}+i}}$; $z_3 = \sqrt[6]{\frac{1+i}{\sqrt{3}-i}}$ which of the following holds good?

(A) $\sum |Z_1|^2 = \frac{3}{2}$ (B) $|Z_1|^4 + |Z_2|^4 = |Z_3|^{-8}$

(C) $\sum |Z_1|^3 + |Z_2|^3 = |Z_3|^{-6}$ (D) $|Z_1|^4 + |Z_2|^4 = |Z_3|^8$

MCN174

15. If z is a complex number satisfying the equation $|z+i| + |z-i| = 8$, on the complex plane then maximum value of $|z|$ is

(A) 2 (B) 4 (C) 6 (D) 8

MCN175

For More Join: @IITJEE_Advanced

16. Let Z be a complex number satisfying the equation $(Z^3 + 3)^2 = -16$ then $|Z|$ has the value equal to
(A) $5^{1/2}$ (B) $5^{1/3}$ (C) $5^{2/3}$ (D) 5

MCN176

17. If z_1, z_2, z_3 are 3 distinct complex numbers such that $\frac{3}{|z_2 - z_3|} = \frac{4}{|z_3 - z_1|} = \frac{5}{|z_1 - z_2|}$, then the value of $\frac{9}{z_2 - z_3} + \frac{16}{z_3 - z_1} + \frac{25}{z_1 - z_2}$ equals
(A) 0 (B) 3 (C) 4 (D) 5

MCN177

18. Consider two complex numbers α and β as $\alpha = \left(\frac{a+bi}{a-bi}\right)^2 + \left(\frac{a-bi}{a+bi}\right)^2$, where $a, b \in R$ and $\beta = \frac{z-1}{z+1}$, $z \neq \pm 1$ where $|z| = 1$, then
(A) Both α and β are purely real
(B) Both α and β are purely imaginary
(C) α is purely real and β is purely imaginary
(D) β is purely real and α is purely imaginary

MCN178

19. Let Z is complex satisfying the equation $z^2 - (3 + i)z + m + 2i = 0$, where $m \in R$. Suppose the equation has a real root. The additive inverse of non real root, is
(A) $1 - i$ (B) $1 + i$ (C) $-1 - i$ (D) -2

MCN179

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20. The minimum value of $|z - 1 + 2i| + |4i - 3 - z|$ is
(A) $\sqrt{5}$ (B) 5 (C) $2\sqrt{13}$ (D) $\sqrt{15}$

MCN180

21. Identify the incorrect statement.
(A) no non zero complex number z satisfies the equation, $\bar{z} = -4z$
(B) $\bar{z} = z$ implies that z is purely real
(C) If z is purely imaginary, then $\bar{z} + z = 0$.
(D) if z_1, z_2 are the roots of the quadratic equation $az^2 + bz + c = 0$ such that Im $(z_1 z_2) \neq 0$ then a, b, c must be real numbers.

MCN181

22. $z_1 = \frac{a}{1-i}$; $z_2 = \frac{b}{2+i}$; $z_3 = a - bi$ for $a, b \in R$ if $z_1 - z_2 = 1$ then the centroid of the triangle formed by the points z_1, z_2, z_3 in the argand's plane is given by
(A) $\frac{1}{9}(1 + 7i)$ (B) $\frac{1}{3}(1 + 7i)$ (C) $\frac{1}{3}(1 - 3i)$ (D) $\frac{1}{9}(1 - 3i)$

MCN182

23. Number of complex numbers z such that $|z| = 1$ and $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$ is
(A) 4 (B) 6 (C) 8 (D) more than 8

MCN183

For More Join: @IITJEE_Advanced

24. If z is a complex number satisfying the equation $|z - (1 + i)|^2 = 2$ and $\omega = \frac{2}{z}$, then the locus traced by 'ω' in the complex plane is

(A) $x - y - 1 = 0$ (B) $x + y - 1 = 0$ (C) $x - y + 1 = 0$ (D) $x + y + 1 = 0$

MCN184

25. A particle starts from a point $z_0 = 1 + i$, where $i = \sqrt{-1}$. It moves horizontally away from origin by 2 units and then vertically away from origin by 3 units to reach a point z_1. From z_1 particle moves $\sqrt{5}$ units in the direction of $2\hat{i} + \hat{j}$ and then it moves through an angle of $\text{cosec}^{-1}\sqrt{2}$ in anticlockwise direction of a circle with centre at origin to reach a point z_2. The $\arg z_2$ is given by

(A) $\sec^{-1}2$ (B) $\cot^{-1}0$ (C) $\sin^{-1}\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)$ (D) $\cos^{-1}\left(\frac{-1}{2}\right)$

MCN185

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For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If z is a complex number which simultaneously satisfies the equations $3|z-12| = 5|z-8i|$ and $|z-4| = |z-8|$ then the Im(z) can be

(A) 15 (B) 16 (C) 17 (D) 8

MCN186

2. Let z_1, z_2, z_3 be non-zero complex numbers satisfying the equation $z^4 = iz$.
Which of the following statement(s) is/are correct?

(A) The complex number having least positive argument is $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$.

(B) $\sum_{k=1}^{3} \text{Amp}(z_k) = \frac{\pi}{2}$

(C) Centroid of the triangle formed by z_1, z_2 and z_3 is $\left(\frac{1}{\sqrt{3}}, \frac{-1}{3}\right)$

(D) Area of triangle formed by z_1, z_2 and z_3 is $\frac{3\sqrt{3}}{2}$

MCN187

3. The locii of a point $P(z)$ in the complex plane satisfying the $\left|z + \frac{1}{z}\right| = 2$ are two circles C_1 and C_2. These circles

(A) have centres on real axis. (B) cut each other orthogonally.
(C) are congruent (D) have exactly two common tangents.

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MCN188

4. Let A and B be two distinct points denoting the complex numbers α and β respectively. A complex number z lies between A and B where $z \neq \alpha$, $z \neq \beta$. Which of the following relation(s) hold good?

(A) $|\alpha - z| + |z - \beta| = |\alpha - \beta|$

(B) $\exists$ a positive real number 't' such that $z = (1-t)\alpha + t\beta$

(C) $\begin{vmatrix} z-\alpha & \bar{z}-\bar{\alpha} \\ \beta-\alpha & \bar{\beta}-\bar{\alpha} \end{vmatrix} = 0$

(D) $\begin{vmatrix} z & \bar{z} & 1 \\ \alpha & \bar{\alpha} & 1 \\ \beta & \bar{\beta} & 1 \end{vmatrix} = 0$

MCN189

5. Equation of a straight line on the complex plane passing through a point P denoting the complex number α and perpendicular to the vector $\overrightarrow{OP}$ where 'O' is the origin can be written as

(A) $\text{Im}\left(\frac{z-\alpha}{\alpha}\right) = 0$ (B) $\text{Re}\left(\frac{z-\alpha}{\alpha}\right) = 0$ (C) $\text{Re}(\bar{\alpha} z) = 0$ (D) $\bar{\alpha} z + \alpha \bar{z} - 2|\alpha|^2 = 0$

MCN190

For More Join: @IITJEE_Advanced

6. If $\alpha_1, \alpha_2, \alpha_3, \ldots\ldots, \alpha_{n-1}$ are the imaginary n^{th} roots of unity then the product $\prod_{r=1}^{n-1}(i-\alpha_r)$ (where $i=\sqrt{-1}$) can take the value equal to

(A) 0 (B) 1 (C) i (D) $(1+i)$

MCN191

7. If the expression $(1+ir)^3$ is of the form of $s(1+i)$ for some real 's' where 'r' is also real and $i=\sqrt{-1}$, then the value of 'r' can be

(A) $\cot\frac{\pi}{8}$ (B) $\sec\pi$ (C) $\tan\frac{\pi}{12}$ (D) $\tan\frac{5\pi}{12}$

MCN192

8. Locus of all points z in argand plane which satisfy $|z^2+1|=|z^2-1|$ is -

(A) same as $\text{Re}(z)=0$ (B) same as $\text{Re}(z^2)=0$

(C) a pair of straight lines (D) a circle with unit radius

MCN193

9. If the biquadratic $x^4+ax^3+bx^2+cx+d=0$ $(a,b,c,d \in R)$ has 4 non real roots, two with sum $3+4i$ and the other two with product $13+i$.

(A) $b=51$ (B) $a=-6$ (C) $c=-70$ (D) $d=170$

MCN194

10. The quadratic equation $z^2+(p+ip')z+q+iq'=0$; where p,p',q,q' are all real.

(A) if the equation has one real root then $q'^2-pp'q'+qp'^2=0$.

(B) if the equation has two equal roots then $pp'=2q'$.

(C) if the equation has two equal roots then $p^2-p'^2=4q$

(D) if the equation has one real root then $p'^2-pp'q'+q'^2=0$.

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MCN195

11. The value of i^n+i^{-n}, for $i=\sqrt{-1}$ and $n \in I$ is :

(A) $\frac{2^n}{(1-i)^{2n}}+\frac{(1+i)^{2n}}{2^n}$ (B) $\frac{(1+i)^{2n}}{2^n}+\frac{(1-i)^{2n}}{2^n}$

(C) $\frac{(1+i)^{2n}}{2^n}+\frac{2^n}{(1-i)^{2n}}$ (D) $\frac{2^n}{(1+i)^{2n}}+\frac{2^n}{(1-i)^{2n}}$

MCN196

12. If amp $(z_1z_2)=0$ and $|z_1|=|z_2|=1$, then

(A) $z_1+z_2=0$ (B) $z_1z_2=1$ (C) $z_1=\overline{z}_2$ (D) $z_1=z_2$

MCN197

13. Let z_1 and z_2 are two complex numbers such that $(1-i)z_1=2z_2$ and $\arg(z_1z_2)=\frac{\pi}{2}$, then $\arg(z_2)$ is equal to

(A) $3\pi/8$ (B) $\pi/8$ (C) $5\pi/8$ (D) $-7\pi/8$

MCN198

14. a, b, c are real numbers in the polynomial, $P(z)=2z^4+az^3+bz^2+cz+3$. If two roots of the equation $P(z)=0$ are 2 and i. Then which of the following are true.

(A) $a=-\frac{11}{2}$ (B) $b=5$ (C) $c=-\frac{11}{2}$ (D) $a=-11$

MCN199

For More Join: @IITJEE_Advanced

15. If $Z = \frac{(1+i)(1+2i)(1+3i).....(1+ni)}{(1-i)(2-i)(3-i).....(n-i)}$, $n \in N$ then principal argument of Z can be

(A) 0 (B) $\frac{\pi}{2}$ (C) $-\frac{\pi}{2}$ (D) π

MCN200

COMPREHENSION TYPE QUESTIONS

Paragraph for question nos. 16 to 18

Consider a complex number $w = \frac{z-i}{2z+1}$, where $z = x + iy$ and $x, y \in R$.

16. If the complex number w is purely imaginary then locus of z is -

(A) a straight line

(B) a circle with centre $\left(-\frac{1}{4}, \frac{1}{2}\right)$ and radius $\frac{\sqrt{5}}{4}$.

(C) a circle with centre $\left(\frac{1}{4}, -\frac{1}{2}\right)$ and passing through origin.

(D) neither a circle nor a straight line.

MCN201

17. If the complex number w is purely real then locus of z is

(A) a straight line passing through origin

(B) a straight line with gradient 3 and y intercept (–1)

(C) a straight line with gradient 2 and y intercept 1.

(D) none

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MCN202

18. If $|w| = 1$ then the locus of $P(z)$ is

(A) a point circle (B) an imaginary circle

(C) a real circle (D) not a circle.

MCN203

Paragraph for question nos. 19 to 20

$ABCD$ is a rhombus. Its diagonals AC and BD intersect at the point M and satisfy $BD = 2AC$. Let the points D and M represent complex numbers $1 + i$ and $2 - i$ respectively.

If θ is arbitrary real, then $z = re^{i\theta}$, $R_1 \le r \le R_2$ lies in annular region formed by concentric circles $|z| = R_1$, $|z| = R_2$.

19. A possible representation of point A is

(A) $3 - \frac{i}{2}$ (B) $1 - \frac{3i}{2}$ (C) $1 + \frac{3}{2}i$ (D) $3 - \frac{3}{2}i$

MCN204

20. If z is any point on segment DM then $w = e^{iz}$ lies in annular region formed by concentric circles.

(A) $|w|_{min} = 1$, $|w|_{max} = 2$ (B) $|w|_{min} = \frac{1}{e}$, $|w|_{max} = e$

(C) $|w|_{min} = \frac{1}{e^2}$, $|w|_{max} = e^2$ (D) $|w|_{min} = \frac{1}{2}$, $|w|_{max} = e$

MCN205

For More Join: @IITJEE_Advanced

Paragraph for question nos. 21 to 23

Let A, B, C be three sets of complex numbers as defined below.

$A = \{z : |z+1| \le 2 + \text{Re}(z)\}$, $B = \{z : |z-1| \ge 1\}$ and $C = \left\{z : \left|\frac{z-1}{z+1}\right| \ge 1\right\}$

21. The number of point(s) having integral coordinates in the region $A \cap B \cap C$ is

MCN206

22. The area of region bounded by $A \cap B \cap C$ is $2\sqrt{\lambda}$ then λ is

MCN207

23. The real part of the complex number in the region $A \cap B \cap C$ and having maximum amplitude can be

MCN208

MATCHING LIST TYPE QUESTION

24. Match the List

List - I (Complex number Z)		**List - II (Principal argument of Z)**	
(A)	$z = \dfrac{(1+i)^5\left(1+\sqrt{3}i\right)^2}{-2i\left(-\sqrt{3}+i\right)}$.	(P)	π
(B)	$z = \sin\dfrac{6\pi}{5} + i\left(1+\cos\dfrac{6\pi}{5}\right)$ is	(Q)	$-\dfrac{7\pi}{18}$
(C)	$z = 1+\cos\left(\dfrac{11\pi}{9}\right)+i\sin\left(\dfrac{11\pi}{9}\right)$	(R)	$\dfrac{9\pi}{10}$
(D)	$z = \sin x \sin(x-60)\sin(x+60)$ where $x \in \left(0, \dfrac{\pi}{3}\right)$ and $x \in R$	(S)	$-\dfrac{5\pi}{12}$
		(T)	0

(A) (A) → (P), (B) → (R), (C) → (S), (D) → (Q)
(B) (A) → (S), (B) → (R), (C) → (Q), (D) → (P)
(C) (A) → (R), (B) → (Q), (C) → (S), (D) → (P)
(D) (A) → (Q), (B) → (S), (C) → (R), (D) → (P)

MCN209

MATRIX MATCH TYPE QUESTION

25. Match the equation in z, in **Column-I** with the corresponding values of arg(z) in **Column-II**.

Column-I (equations in z)		**Column-II (principal value of arg (z))**	
(A)	$z^2 - z + 1 = 0$	(P)	$-2\pi/3$
(B)	$z^2 + z + 1 = 0$	(Q)	$-\pi/3$
(C)	$2z^2 + 1 + i\sqrt{3} = 0$	(R)	$\pi/3$
(D)	$2z^2 + 1 - i\sqrt{3} = 0$	(S)	$2\pi/3$

MCN210

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EXERCISE (S)

1. If a and b are positive integer such that $N = (a + ib)^3 - 107i$ is a positive integer then find the value of $\frac{N}{2}$.

MCN211

2. Let z, w be complex numbers such that $\bar{z} + i\bar{w} = 0$ and arg $zw = \pi$. If $Re(z) < 0$ and principal arg $z = \frac{a\pi}{b}$ then find the value of $a + b$. (where a & b are co-prime natural numbers)

MCN212

3. If $x = 9^{1/3}\ 9^{1/9}\ 9^{1/27} \ldots \infty$, $y = 4^{1/3}\ 4^{-1/9}\ 4^{1/27} \ldots \infty$, and $z = \sum_{r=1}^{\infty}(1+i)^{-r}$ and principal argument of $P = (x + yz)$ is $-\tan^{-1}\left(\frac{\sqrt{a}}{b}\right)$ then determine $a^2 + b^2$. (where a & b are co-prime natural numbers)

MCN213

4. $z_1, z_2 \in c$ and $z_1^2 + z_2^2 \in R$, $z_1(z_1^2 - 3z_2^2) = 2$, $z_2(3z_1^2 - z_2^2) = 11$. If $z_1^2 + z_2^2 = \lambda$ then determine λ^2

MCN214

5. Let $|z| = 2$ and $w = \frac{z+1}{z-1}$ where $z, w \in C$ (where C is the set of complex numbers), then find product of maximum and minimum value of $|w|$.

MCN215

6. A function 'f' is defined by $f(z) = (4 + i)z^2 + \alpha z + \gamma$ for all complex number z, where α and γ are complex numbers if $f(1)$ and $f(i)$ are both real and the smallest possible values of $|\alpha| + |\gamma|$ is p then determine p.

MCN216

7. If z and ω are two non-zero complex numbers such that $|z\omega| = 1$, and arg (z) – arg$(\omega) = \frac{\pi}{2}$, then find the value of $\frac{5i\bar{z}\omega}{4}$

MCN217

8. If $a_1, a_2, a_3 \ldots a_n, A_1, A_2, A_3 \ldots A_n, k$ are all real numbers and number of imaginary roots of the equation $\frac{A_1^2}{x - a_1} + \frac{A_2^2}{x - a_2} + \ldots\ldots + \frac{A_n^2}{x - a_n} = k$ is α. Then find the value of $\frac{\alpha + 15}{10}$.

MCN218

9. If a variable circle S touches $S_1 : |z - z_1| = 7$ internally and $S_2 : |z - z_2| = 4$ externally while the curves S_1 & S_2 touch internally to each other, $(z_1 \neq z_2)$. If the eccentricity of the locus of the centre of the curve S is 'e' find the value of e.

MCN219

10. Area of the region formed by $|z| \leq 4$ & $-\frac{\pi}{2} \leq \arg z \leq \frac{\pi}{3}$ on the Argand diagram is expressed in the form of $\frac{a\pi}{b}$. Then find the value of $\frac{a}{b}$ (where a & b are co-prime natural number)

MCN220

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EXERCISE - JEE (Main) PYQ

1. Let $z \in C$ be such that $|z| < 1$. If $\omega = \frac{5+3z}{5(1-z)}$, then - **[JEE (Main) 2019]**

(1) $5\text{Im}(\omega) < 1$ (2) $4\text{Im}(\omega) > 5$ (3) $5\text{Re}(\omega) > 1$ (4) $5\text{Re}(\omega) > 4$

MCN035

2. If $\text{Re}\left(\frac{z-1}{2z+i}\right) = 1$, where $z = x + iy$, then the point (x, y) lies on a : **[JEE (Main) 2020]**

(1) circle whose centre is at $\left(-\frac{1}{2}, -\frac{3}{2}\right)$ (2) circle whose diameter is $\frac{\sqrt{5}}{2}$

(3) straight line whose slope is $\frac{3}{2}$ (4) straight line whose slope is $-\frac{2}{3}$

MCN036

3. The value of $\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$ is : **[JEE (Main) 2020]**

(1) $\frac{1}{2}(\sqrt{3}-i)$ (2) $-\frac{1}{2}(\sqrt{3}-i)$ (3) $-\frac{1}{2}(1-i\sqrt{3})$ (4) $\frac{1}{2}(1-i\sqrt{3})$

MCN037

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4. If a and b are real numbers such that $(2+\alpha)^4 = a + b\alpha$, where $\alpha = \frac{-1+i\sqrt{3}}{2}$, then $a + b$ is equal to : **[JEE (Main) 2020]**

(1) 57 (2) 33 (3) 24 (4) 9

MCN038

5. The value of $\left(\frac{-1+i\sqrt{3}}{1-i}\right)^{30}$ is : **[JEE (Main) 2020]**

(1) $2^{15}\, i$ (2) -2^{15} (3) $-2^{15}\, i$ (4) 6^5

MCN039

6. Let $i = \sqrt{-1}$. If $\frac{(-1+i\sqrt{3})^{21}}{(1-i)^{24}} + \frac{(1+i\sqrt{3})^{21}}{(1+i)^{24}} = k$, and $n = [|k|]$ be the greatest integral part of $|k|$. Then $\sum_{j=0}^{n+5}(j+5)^2 - \sum_{j=0}^{n+5}(j+5)$ is equal to. **[JEE (Main) 2021]**

MCN040

7. Let the lines $(2-i)z = (2+i)\bar{z}$ and $(2+i)z + (i-2)\bar{z} - 4i = 0$, (here $i^2 = -1$) be normal to a circle C. If the line $iz + \bar{z} + 1 + i = 0$ is tangent to this circle C, then its radius is: **[JEE (Main) 2021]**

(1) $\frac{3}{\sqrt{2}}$ (2) $\frac{1}{2\sqrt{2}}$ (3) $3\sqrt{2}$ (4) $\frac{3}{2\sqrt{2}}$

MCN041

For More Join: @IITJEE_Advanced

8. The least value of $|z|$ where z is complex number which satisfies the inequality

$\exp\left(\frac{(|z|+3)(|z|-1)}{||z|+1|}\log_e 2\right) \geq \log_{\sqrt{2}}\left|5\sqrt{7}+9i\right|, i=\sqrt{-1}$, is equal to : **[JEE (Main) 2021]**

(1) 3 (2) $\sqrt{5}$ (3) 2 (4) 8

MCN042

9. Let S_1, S_2 and S_3 be three sets defined as

$S_1 = \{z \in \mathbb{C} : |z-1| \leq \sqrt{2}\}$

$S_2 = \{z \in \mathbb{C} : Re((1-i)z) \geq 1\}$

$S_3 = \{z \in \mathbb{C} : Im(z) \leq 1\}$

Then the set $S_1 \cap S_2 \cap S_3$ **[JEE (Main) 2021]**

(1) is a singleton (2) has exactly two elements

(3) has infinitely many elements (4) has exactly three elements

MCN043

10. If z and ω are two complex numbers such that $|z\omega| = 1$ and $\arg(z) - \arg(\omega) = \frac{3\pi}{2}$, then $\arg\left(\frac{1-2\bar{z}\omega}{1+3\bar{z}\omega}\right)$ is : (Here $\arg(z)$ denotes the principal argument of complex number z)

[JEE (Main) 2021]

(1) $\frac{\pi}{4}$ (2) $-\frac{3\pi}{4}$ (3) $-\frac{\pi}{4}$ (4) $\frac{3\pi}{4}$

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MCN044

11. Let $\mathbb{C}$ be the set of all complex numbers. Let

$S_1 = \{z \in \mathbb{C} : |z-2| \leq 1\}$ and

$S_2 = \{z \in \mathbb{C} : z(1+i) + \bar{z}(1-i) \geq 4\}$.

Then, the maximum value of $\left|z-\frac{5}{2}\right|^2$ for $z \in S_1 \cap S_2$ is equal to : **[JEE (Main) 2021]**

(1) $\frac{3+2\sqrt{2}}{4}$ (2) $\frac{5+2\sqrt{2}}{2}$ (3) $\frac{3+2\sqrt{2}}{2}$ (4) $\frac{5+2\sqrt{2}}{4}$

MCN045

12. Let $z=\frac{1-i\sqrt{3}}{2}, i=\sqrt{-1}$ then the value of

$21+\left(z+\frac{1}{z}\right)^3+\left(z^2+\frac{1}{z^2}\right)^3+\left(z^3+\frac{1}{z^3}\right)^3+\ldots+\left(z^{21}+\frac{1}{z^{21}}\right)^3$ is. **[JEE (Main) 2021]**

MCN046

13. If $(\sqrt{3}+i)^{100} = 2^{99}(p+iq)$, then p and q are roots of the equation : **[JEE (Main) 2021]**

(1) $x^2-(\sqrt{3}-1)x-\sqrt{3}=0$ (2) $x^2+(\sqrt{3}+1)x+\sqrt{3}=0$

(3) $x^2+(\sqrt{3}-1)x-\sqrt{3}=0$ (4) $x^2-(\sqrt{3}+1)x+\sqrt{3}=0$

MCN047

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14. If $S=\left\{z\in\mathbb{C}:\frac{z-i}{z+2i}\in\mathbb{R}\right\}$, then : **[JEE (Main) 2021]**

(1) S contains exactly two elements (2) S contains only one element

(3) S is a circle in the complex plane (4) S is a straight line in the complex plane

MCN048

15. If $a_r=\cos\frac{2r\pi}{9}+i\sin\frac{2r\pi}{9}$, $r=1,2,3,....i=\sqrt{-1}$ then the determinant $\begin{vmatrix}a_1 & a_2 & a_3\\ a_4 & a_5 & a_6\\ a_7 & a_8 & a_9\end{vmatrix}$ is equal to

[JEE (Main) 2021]

(1) $a_2a_6-a_4a_8$ (2) a_9 (3) $a_1a_9-a_3a_7$ (4) a_5

MCN049

16. For $z\in\mathbb{C}$ if the minimum value of $\left(|z-3\sqrt{2}|+|z-p\sqrt{2}i|\right)$ is $5\sqrt{2}$, then a value of p is _____

[JEE (Main) 2022]

(1) 3 (2) $\frac{7}{2}$ (3) 4 (4) $\frac{9}{2}$

MCN050

17. Let S be the set of all (α,β), $\pi<\alpha,\beta<2\pi$, for which the complex number $\frac{1-i\sin\alpha}{1+2i\sin\alpha}$ is purely imaginary and $\frac{1+i\cos\beta}{1-2i\cos\beta}$ is purely real. Let $Z_{\alpha\beta}=\sin2\alpha+i\cos2\beta,(\alpha,\beta)\in S$. Then $\sum_{(\alpha,\beta)\in S}\left(iZ_{\alpha\beta}+\frac{1}{i\overline{Z}_{\alpha\beta}}\right)$ is equal to : **[JEE (Main) 2022]**

(1) 3 (2) $3i$ (3) 1 (4) $2-i$

MCN051

18. Let $S=\{z=x+iy:|z-1+i|\geq|z|,|z|<2,|z+i|=|z-1|\}$. Then the set of all values of x, for which $w=2x+iy\in S$ for some $y\in\mathbb{R}$, is **[JEE (Main) 2022]**

(1) $\left(-\sqrt{2},\frac{1}{2\sqrt{2}}\right]$ (2) $\left(-\frac{1}{\sqrt{2}},\frac{1}{4}\right]$ (3) $\left(-\sqrt{2},\frac{1}{2}\right]$ (4) $\left(-\frac{1}{\sqrt{2}},\frac{1}{2\sqrt{2}}\right]$

MCN052

19. Let $S=\left\{z\in\mathbb{C}:|z-3|\leq1\text{ and }z(4+3i)+\overline{z}(4-3i)\leq24\right\}$. If $\alpha+i\beta$ is the point in S which is closest to $4i$, then $25(\alpha+\beta)$ is equal to ____. **[JEE (Main) 2022]**

MCN053

20. Let a circle C in complex plane pass through the points $z_1=3+4i$, $z_2=4+3i$ and $z_3=5i$. If $z(\neq z_1)$ is a point on C such that the line through z and z_1 is perpendicular to the line through z_2 and z_3, then $\arg(z)$ is equal to : **[JEE (Main) 2022]**

(1) $\tan^{-1}\left(\frac{2}{\sqrt{5}}\right)-\pi$ (2) $\tan^{-1}\left(\frac{24}{7}\right)-\pi$ (3) $\tan^{-1}(3)-\pi$ (4) $\tan^{-1}\left(\frac{3}{4}\right)-\pi$

MCN054

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21. The number of points of intersection of $|z-(4+3i)|=2$ and $|z|+|z-4|=6$, $z \in C$ is :

[JEE (Main) 2023]

(1) 0 (2) 1 (3) 2 (4) 3

MCN055

22. Let α,β,γ be the three roots of the equation $x^3+bx+c=0$. If $\beta\gamma=1=-\alpha$, then $b^3+2c^3-3\alpha^3-6\beta^3-8\gamma^3$ is equal to **[JEE (Main) 2023]**

(1) 21 (2) $\frac{169}{8}$ (3) 19 (4) $\frac{155}{8}$

MCN056

23. Let the complex number $z = x + iy$ be such that $\frac{2z-3i}{2z+i}$ is purely imaginary. If $x + y^2 = 0$, then $y^4 + y^2 - y$ is equal to : **[JEE (Main) 2023]**

(1) $\frac{3}{2}$ (2) $\frac{4}{3}$ (3) $\frac{2}{3}$ (4) $\frac{3}{4}$

MCN057

24. For $a \in C$, let $A=\{z \in C : \text{Re}(a+\bar{z}) > Im(\bar{a}+z)\}$ and $B=\{z \in C : \text{Re}(a+\bar{z}) < Im(\bar{a}+z)\}$. Then among the two statements : **[JEE (Main) 2023]**

(S1) : If $Re(A), Im(A) > 0$, then the set A contains all the real numbers

(S2) : If $Re(A), Im(A) < 0$, then the set B contains all the real numbers,

(1) Only (S1) is true (2) both are false (3) Only (S2) is true (4) Both are true

MCN058

25. Let $z_1 = 2+3i$ and $z_2 = 3+4i$. The set $S=\{z \in C : |z-z_1|^2-|z-z_2|^2=|z_1-z_2|^2\}$ represents a

[JEE (Main) 2023]

(1) straight line with sum of its intercepts on the coordinate axes equals 14

(2) hyperbola with the length of the transverse axis 7

(3) straight line with the sum of its intercepts on the coordinate axes equals –18

(4) hyperbola with eccentricity 2

MCN059

26. Let $\alpha = 8 - 14i$, $A=\left\{z \in \mathbb{C} : \frac{\alpha z-\bar{\alpha}\bar{z}}{z^2-(\bar{z})^2-112i}=1\right\}$ and $B=\{z \in \mathbb{C} : |z+3i|=4\}$.

Then $\sum_{z \in A \cap B} (\text{Re}\, z - \text{Im}\, z)$ is equal to ____. **[JEE (Main) 2023]**

MCN060

27. The area (in sq. units) of the region $S=\{z \in \mathbb{C}; |z-1| \le 2; (z+\bar{z})+i(z-\bar{z}) \le 2, Im(z) \ge 0\}$ is

[JEE (Main) 2024]

(1) $\frac{7\pi}{3}$ (2) $\frac{3\pi}{2}$ (3) $\frac{17\pi}{8}$ (4) $\frac{7\pi}{4}$

MCN221

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28. Consider the following two statements :

Statement I : For any two non-zero complex numbers z_1, z_2

$\left(|z_1|+|z_2|\right)\left|\frac{z_1}{|z_1|}+\frac{z_2}{|z_2|}\right| \le 2\left(|z_1|+|z_2|\right)$ and

Statement II : If x, y, z are three distinct complex numbers and a, b, c are three positive real numbers such that $\frac{a}{|y-z|}=\frac{b}{|z-x|}=\frac{c}{|x-y|}$, then

$\frac{a^2}{y-z}+\frac{b^2}{z-x}+\frac{c^2}{x-y}=1$.

Between the above two statements, **[JEE (Main) 2024]**

(1) both Statement I and Statement II are incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Statement I is correct but Statement II is incorrect.

(4) both Statement I and Statement II are correct.

MCN222

29. If z_1, z_2 are two distinct complex number such that $\left|\frac{z_1-2z_2}{\frac{1}{2}-z_1\bar{z}_2}\right|=2$, then **[JEE (Main) 2024]**

(1) either z_1 lies on a circle of radius 1 or z_2 lies on a circle of radius $\frac{1}{2}$

(2) either z_1 lies on a circle of radius $\frac{1}{2}$ or z_2 lies on a circle of radius 1.

(3) z_1 lies on a circle of radius $\frac{1}{2}$ and z_2 lies on a circle of radius 1.

(4) both z_1 and z_2 lie on the same circle.

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MCN223

30. Let x_1, x_2, x_3, x_4 be the solution of the equation $4x^4 + 8x^3 - 17x^2 - 12x + 9 = 0$ and $\left(4+x_1^2\right)\left(4+x_2^2\right)\left(4+x_3^2\right)\left(4+x_4^2\right)=\frac{125}{16}m$. Then the value of m is _____. **[JEE (Main) 2024]**

MCN224

31. If the set $R = \{(a, b) ; a + 5b = 42, a, b \in \mathbb{N}\}$ has m elements and $\sum_{n=1}^{m}\left(1+i^{n!}\right)=x+iy$, where $I=\sqrt{-1}$, then the value of $m + x + y$ is : **[JEE (Main) 2024]**

(1) 8 (2) 12 (3) 4 (4) 5

MCN225

32. Let $S=\{z \in C:|z-1|=1$ and $\left(\sqrt{2}-1\right)(z+\bar{z})-i(z-\bar{z})=2\sqrt{2}\}$. Let $z_1, z_2 \in S$ be such that $|z_1|=\max_{z\in s}|z|$ and $|z_2|=\min_{z\in s}|z|$. Then $\left|\sqrt{2}z_1-z_2\right|^2$ equals : **[JEE (Main) 2024]**

(1) 1 (2) 4 (3) 3 (4) 2

MCN226

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33. Let the complex numbers α and $\frac{1}{\bar{\alpha}}$ lie on the circles $|z-z_0|^2=4$ and $|z-z_0|^2=16$ respectively, where $z_0=1+i$. Then, the value of $100|\alpha|^2$ is..... **[JEE (Main) 2024]**

MCN227

34. Let α, β be the roots of the equation $x^2-\sqrt{6}x+3=0$ such that Im (α) > Im (β). Let a, b be integers not divisible by 3 and n be a natural number such that $\frac{\alpha^{99}}{\beta}+\alpha^{98}=3^n(a+ib), i=\sqrt{-1}$. Then $n + a + b$ is equal to _____. **[JEE (Main) 2024]**

MCN228

35. If α denotes the number of solutions of $|1-i|^x = 2^x$ and $\beta=\left(\frac{|z|}{\arg(z)}\right)$, where $z=\frac{\pi}{4}(1+i)^4\left(\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i}+\frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right)$, $i=\sqrt{-1}$, then the distance of the point (α, β) from the line $4x-3y=7$ is ____ **[JEE (Main) 2024]**

MCN229

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EXERCISE - JEE (Advanced) PYQ

1. Let ω be a complex cube root of unity with $\omega \neq 1$ and $P = [p_{ij}]$ be a $n \times n$ matrix with $p_{ij} = \omega^{i+j}$. Then $P^2 \neq 0$, when n = **[JEE (Advanced) 2013]**

(A) 57 (B) 55 (C) 58 (D) 56

MCN104

2. Let $w = \frac{\sqrt{3}+i}{2}$ and $P = \{w^n : n = 1, 2, 3, ...\}$. Further $H_1 = \left\{z \in C : \text{Re}\, z > \frac{1}{2}\right\}$ and , $H_2 = \left\{z \in C : \text{Re}\, z < \frac{-1}{2}\right\}$ where C is the set of all complex numbers. If $z_1 \in P \cap H_1$, $z_2 \in P \cap H_2$ and O represents the origin, then $\angle z_1 O z_2$ = **[JEE (Advanced) 2013]**

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{6}$ (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{6}$

MCN105

Paragraph for Question 3 and 4

Let $S = S_1 \cap S_2 \cap S_3$, where

$S_1 = \{z \in C : |z| < 4\}$, $S_2 = \left\{z \in C : \text{Im}\left[\frac{z-1+\sqrt{3}i}{1-\sqrt{3}i}\right] > 0\right\}$ and $S_3 = \{z \in C : \text{Re}\, z > 0\}$.

3. $\min_{z \in S} |1-3i-z| =$ **[JEE (Advanced) 2013]**

(A) $\frac{2-\sqrt{3}}{2}$ (B) $\frac{2+\sqrt{3}}{2}$ (C) $\frac{3-\sqrt{3}}{2}$ (D) $\frac{3+\sqrt{3}}{2}$

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MCN106

4. Area of S = **[JEE (Advanced) 2013]**

(A) $\frac{10\pi}{3}$ (B) $\frac{20\pi}{3}$ (C) $\frac{16\pi}{3}$ (D) $\frac{32\pi}{3}$

MCN107

5. Let $z_k = \cos\left(\frac{2k\pi}{10}\right) + i\sin\left(\frac{2k\pi}{10}\right)$; k = 1, 2, ... 9. **[JEE (Advanced) 2014]**

List-I	**List-II**
P. For each z_k there exists a z_j such that $z_k \cdot z_j = 1$	1. True
Q. There exists a $k \in \{1, 2, ..., 9\}$ such that $z_1 \cdot z = z_k$ has no solution z in the set of complex numbers.	2. False
R. $\frac{\lvert 1-z_1\rvert\lvert 1-z_2\rvert ... \lvert 1-z_9\rvert}{10}$ equals	3. 1
S. $1 - \sum_{k=1}^{9} \cos\left(\frac{2k\pi}{10}\right)$ equals	4. 2

Codes :

	P	Q	R	S		P	Q	R	S
(A)	1	2	4	3	(B)	2	1	3	4
(C)	1	2	3	4	(D)	2	1	4	3

MCN108

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6.

	Column-I	Column-II
(A)	In $\mathbb{R}^2$, if the magnitude of the projection vector of the vector $\alpha\hat{i}+\beta\hat{j}$ on $\sqrt{3}\hat{i}+\hat{j}$ is $\sqrt{3}$ and if $\alpha=2+\sqrt{3}\beta$, then possible value(s) of $\lvert\alpha\rvert$ is (are)	(P) 1
(B)	Let a and b be real numbers such that the function $f(x)=\begin{cases}-3ax^2-2, & x<1\\ bx+a^2, & x\geq 1\end{cases}$ is differentiable for all $x \in \mathbb{R}$. Then possible value(s) of a is (are)	(Q) 2
(C)	Let $\omega \neq 1$ be a complex cube root of unity. If $(3-3\omega+2\omega^2)^{4n+3}+(2+3\omega-3\omega^2)^{4n+3}+(-3+2\omega+3\omega^2)^{4n+3}=0$ then possible value(s) of n is (are)	(R) 3
(D)	Let the harmonic mean of two positive real number a and b be 4, If q is a positive real number such that $a, 5, q, b$ is an arithmetic progression, then the value(s) of $\lvert q-a\rvert$ is (are)	(S) 4
		(T) 5

[JEE (Advanced) 2015]

MCN109

7. For any integer k, let $\alpha_k=\cos\left(\frac{k\pi}{7}\right)+i\sin\left(\frac{k\pi}{7}\right)$, where $i=\sqrt{-1}$. The value of the expression

$$\frac{\sum_{k=1}^{12}\lvert\alpha_{k+1}-\alpha_k\rvert}{\sum_{k=1}^{3}\lvert\alpha_{4k-1}-\alpha_{4k-2}\rvert}$$ is

[JEE (Advanced) 2015]

MCN110

8. Let $z=\frac{-1+\sqrt{3}i}{2}$, where $i=\sqrt{-1}$, and $r, s \in \{1, 2, 3\}$. Let $P=\begin{bmatrix}(-z)^r & z^{2s}\\ z^{2s} & z^r\end{bmatrix}$ and I be the identity matrix of order 2. Then the total number of ordered pairs (r, s) for which $P^2=-I$ is

[JEE (Advanced) 2016]

MCN079

9. Let $a, b \in \mathbb{R}$ and $a^2+b^2 \neq 0$. Suppose $S=\left\{z\in\mathbb{C}: z=\frac{1}{a+ibt}, t\in\mathbb{R}, t\neq 0\right\}$, where $i=\sqrt{-1}$.

If $z=x+iy$ and $z \in S$, then (x, y) lies on **[JEE (Advanced) 2016]**

(A) the circle with radius $\frac{1}{2a}$ and centre $\left(\frac{1}{2a}, 0\right)$ for $a>0, b\neq 0$

(B) the circle with radius $-\frac{1}{2a}$ and centre $\left(-\frac{1}{2a}, 0\right)$ for $a<0, b\neq 0$

(C) the x-axis for $a\neq 0, b=0$

(D) the y-axis for $a=0, b\neq 0$

MCN080

For More Join: @IITJEE_Advanced

10. Let a, b, x and y be real numbers such that $a-b=1$ and $y \neq 0$. If the complex number $z=x+iy$ satisfies $\text{Im}\left(\frac{az+b}{z+1}\right)=y$, then which of the following is(are) possible value(s) of x ?

[JEE (Advanced) 2017]

(A) $-1-\sqrt{1-y^2}$ (B) $1+\sqrt{1+y^2}$ (C) $1-\sqrt{1+y^2}$ (D) $-1+\sqrt{1-y^2}$

MCN081

11. For a non-zero complex number z, let $\arg(z)$ denotes the principal argument with $-\pi < arg(z) \leq \pi$. Then, which of the following statement(s) is (are) **FALSE** ?

[JEE (Advanced) 2018]

(A) $\arg(-1-i)=\frac{\pi}{4}$, where $i=\sqrt{-1}$

(B) The function $f: \mathbb{R} \to (-\pi, \pi]$, defined by $f(t)=\arg(-1+it)$ for all $t \in \mathbb{R}$, is continuous at all points of $\mathbb{R}$, where $i=\sqrt{-1}$

(C) For any two non-zero complex numbers z_1 and z_2, $\arg\left(\frac{z_1}{z_2}\right)-arg(z_1)+arg(z_2)$ is an integer multiple of 2π

(D) For any three given distinct complex numbers z_1, z_2 and z_3, the locus of the point z satisfying the condition $\arg\left(\frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}\right)=\pi$, lies on a straight line

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MCN082

12. Let s, t, r be the non-zero complex numbers and L be the set of solutions $z=x+iy$ $(x, y \in \mathbb{R}, i=\sqrt{-1})$ of the equation $sz+t\bar{z}+r=0$, where $\bar{z}=x-iy$. Then, which of the following statement(s) is (are) TRUE ?

[JEE (Advanced) 2018]

(A) If L has exactly one element, then $|s| \neq |t|$

(B) If $|s|=|t|$, then L has infinitely many elements

(C) The number of elements in $L \cap \{z: |z-1+i|=5\}$ is at most 2

(D) If L has more than one element, then L has infinitely many elements

MCN083

13. Let S be the set of all complex numbers z satisfying $|z-2+i| \geq \sqrt{5}$. If the complex number z_0 is such that $\frac{1}{|z_0-1|}$ is the maximum of the set $\left\{\frac{1}{|z-1|}: z \in S\right\}$, then the principal argument of $\frac{4-z_0-\bar{z}_0}{z_0-\bar{z}_0+2i}$ is

[JEE (Advanced) 2019]

(A) $\frac{\pi}{4}$ (B) $-\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{2}$

MCN084

14. Let $\omega \neq 1$ be a cube root of unity. Then the minimum of the set $\{|a+b\omega+c\omega^2|^2 : a, b, c$ distinct non-zero integers$\}$ equals

[JEE (Advanced) 2019]

MCN085

For More Join: @IITJEE_Advanced

15. Let S be the set of all complex numbers z satisfying $|z^2 + z + 1| = 1$. Then which of the following statements is/are TRUE ? **[JEE (Advanced) 2020]**

(A) $\left|z+\frac{1}{2}\right| \le \frac{1}{2}$ for all $z \in S$

(B) $|z| \le 2$ for all $z \in S$

(C) $\left|z+\frac{1}{2}\right| \ge \frac{1}{2}$ for all $z \in S$

(D) The set S has exactly four elements

MCN086

16. For a complex number z, let $\text{Re}(z)$ denote the real part of z. Let S be the set of all complex numbers z satisfying $z^4 - |z|^4 = 4iz^2$, where $i = \sqrt{-1}$. Then the minimum possible value of $|z_1 - z_2|^2$, where $z_1, z_2 \in S$ with $\text{Re}(z_1) > 0$ and $\text{Re}(z_2) < 0$, is **[JEE (Advanced) 2020]**

MCN087

17. Let $\theta_1, \theta_2, ..., \theta_{10}$ be positive valued angles (in radian) such that $\theta_1 + \theta_2 + ... + \theta_{10} = 2\pi$. Define the complex numbers $z_1 = e^{i\theta_1}$, $z_k = z_{k-1}e^{i\theta_k}$ for $k = 2, 3, ..., 10$, where $i = \sqrt{-1}$. Consider the statements P and Q given below : **[JEE (Advanced) 2021]**

$P : |z_2 - z_1| + |z_3 - z_2| + ... + |z_{10} - z_9| + |z_1 - z_{10}| \le 2\pi$

$Q : \left|z_2^2 - z_1^2\right| + \left|z_3^2 - z_2^2\right| + ... + \left|z_{10}^2 - z_9^2\right| + \left|z_1^2 - z_{10}^2\right| \le 4\pi$

Then,

(A) P is **TRUE** and Q is **FALSE**

(B) Q is **TRUE** and P is **FALSE**

(C) both P and Q are **TRUE**

(D) both P and Q are **FALSE**

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MCN088

18. For any complex number $w = c + id$, let $\arg(w) \in (-\pi, \pi]$, where $i = \sqrt{-1}$. Let α and β be real numbers such that for all complex numbers $z = x + iy$ satisfying $\arg\left(\frac{z+\alpha}{z+\beta}\right) = \frac{\pi}{4}$, the ordered pair (x, y) lies on the circle $x^2 + y^2 + 5x - 3y + 4 = 0$. Then which of the following statements is (are) **TRUE** ? **[JEE (Advanced) 2021]**

(A) $\alpha = -1$ (B) $\alpha\beta = 4$ (C) $\alpha\beta = -4$ (D) $\beta = 4$

MCN089

19. Let z be a complex number with non-zero imaginary part. If $\frac{2+3z+4z^2}{2-3z+4z^2}$ is a real number, then the value of $|z|^2$ is _____. **[JEE (Advanced) 2022]**

MCN090

20. Let $\bar{z}$ denote the complex conjugate of a complex number z and let $i = \sqrt{-1}$. In the set of complex numbers, the number of distinct roots of the equation $\bar{z} - z^2 = i(\bar{z} + z^2)$ is. **[JEE (Advanced) 2022]**

MCN091

For More Join: @IITJEE_Advanced

21. Let $\bar{z}$ denote the complex conjugate of a complex number z. If z is a non-zero complex number for which both real and imaginary parts of $(\bar{z})^2+\frac{1}{z^2}$ are integers, then which of the following is/are possible value(s) of $|z|$? **[JEE (Advanced) 2022]**

(A) $\left(\frac{43+3\sqrt{205}}{2}\right)^{\frac{1}{4}}$ (B) $\left(\frac{7+\sqrt{33}}{4}\right)^{\frac{1}{4}}$ (C) $\left(\frac{9+\sqrt{65}}{4}\right)^{\frac{1}{4}}$ (D) $\left(\frac{7+\sqrt{13}}{6}\right)^{\frac{1}{4}}$

MCN092

22. Let $A=\left\{\frac{1967+1686i\sin\theta}{7-3i\cos\theta}:\theta\in\mathbb{R}\right\}$. If A contains exactly one positive integer n, then the value of n is **[JEE (Advanced) 2023]**

MCN093

23. Let z be complex number satisfying $|z|^3+2z^2+4\bar{z}-8=0$, where $\bar{z}$ denotes the complex conjugate of z. Let the imaginary part of z be nonzero. **[JEE (Advanced) 2023]**

Match each entry in **List-I** to the correct entries in **List-II**.

	List-I		**List-II**
(P)	$\|z\|^2$ is equal to	(1)	12
(Q)	$\|z-\bar{z}\|^2$ is equal to	(2)	4
(R)	$\|z\|^2+\|z+\bar{z}\|^2$ is equal to	(3)	8
(S)	$\|z+1\|^2$ is equal to	(4)	10
		(5)	7

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The correct option is :

(A) (P) → (1) (Q) → (3) (R) → (5) (S) → (4)

(B) (P) → (2) (Q) → (1) (R) → (3) (S) → (5)

(C) (P) → (2) (Q) → (4) (R) → (5) (S) → (1)

(D) (P) → (2) (Q) → (3) (R) → (5) (S) → (4)

MCN094

24. Let $A_1, A_2, A_3, \ldots\ldots, A_8$ be the vertices of a regular octagon that lie on a circle of radius 2. Let P be a point on the circle and let Pa_i denote the distance between the points P and A_i for i = 1,2,....,8. If P varies over the circle, then the maximum value of the product $PA_1 \cdot PA_2 \cdot\ldots\ldots\cdot PA_8$, is **[JEE (Advanced) 2023]**

MCN095

25. Let $f(x)=x^4+ax^3+bx^2+c$ be a polynomial with real coefficients such that $f(1)$ = –9. Suppose that $i\sqrt{3}$ is a root of the equation $4x^3+3ax^2+2bx=0$, where $i=\sqrt{-1}$. If α_1, α_2, α_3, and α_4 are all the roots of the equation $f(x)=0$, then $|\alpha_1|^2+|\alpha_2|^2+|\alpha_3|^2+|\alpha_4|^2$ is equal to ..... **[JEE (Advanced) 2024]**

MCN230

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	C	D	B	C	B	D	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	A	B	B	B	A	C	C	C
Que.	21	22	23	24	25					
Ans.	D	A	C	A	B					

EXERCISE – O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C,D	A,B	B,C,D	A,B,C,D	B,D	A,B,C,D	B,C,D	B,C	A,B,C,D	A,B,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,D	B,C	B,D	A,B,C	A,B,C,D	B	C	C	A,B	B,D
Que.	21	22	23	24	25					
Ans.	5.00	3.00	–1.50	B	A → Q,R ; B → P,S ; C → Q,S ; D → P,R					

EXERCISE - S

1. 99 **2.** 7 **3.** 13 **4.** 25 **5.** 1 **6.** 1.41 or 1.42
7. 1.25 **8.** 1.50 **9.** 0.27 or 0.28 **10.** 6.66 or 6.67

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	2	4	3	310	4	1	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	13	1	4	3	3	3	2	80	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	3	4	2	1	14	2	3	1	221
Que.	31	32	33	34	35					
Ans.	2	4	20	49	3					

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6				
Ans.	B,C,D	C,D	C	B	C	A → P,Q; B → P,Q; C → P,Q,S,T; D → Q,T				
Que.	7	8	9	10	11	12	13	14	15	16
Ans.	4	1	A,C,D	A,D	A,B,D	A,C,D	B	3.00	B,C	8.00
Que.	17	18	19	20	21	22	23	24	25	
Ans.	C	B,D	0.50	4.00	A	281	B	512	20	

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